



Homework 1

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EXOPLANETS

Due to September 04, 2023

Problem 1: transiting exoplanets [5 points]

Erect a Cartesian reference frame with $\hat{\mathbf{z}}$ pointing towards the observer using the orbital elements seen in class: Ω and I to orient the orbital planet and e, f, ω to describe the elliptical motion in that plane. Define $\mathbf{r} = (x, y, z)$ with magnitude r .

a) [1 point] In this frame, what is the condition on (x, y, z) for a planet to transit?

Considering the position of the planet is given by $\mathbf{r} = (x, y, z)$, then we have the following area restriction:

$$x^2 + y^2 \leq R_*^2 \quad (1)$$

b) [1 point] Write the condition in terms of I and r . Check that the condition is the same seen in class for a circular orbit ($r = a$).

In order to transform coordinates, we use the transformations seen in class, given by:

$$x = r (\cos \Omega \cos(f + \omega) - \sin \Omega \cos I \sin(f + \omega)) \quad (2)$$

$$y = r (\sin \Omega \cos(f + \omega) + \cos \Omega \cos I \sin(f + \omega)) \quad (3)$$

$$z = r \sin I \sin(f + \omega) \quad (4)$$

Replacing x and y values in the restriction obtained in part a) we get that

$$R_*^2 \geq r^2 [\cos^2(f + \omega) + \cos^2 I \sin^2(f + \omega)] \quad (5)$$

Now considering a circular orbit, the ω value is equal to 0. So the expression becomes:

$$R_*^2 \geq a^2 [\cos^2(f) + \cos^2 I \sin^2(f)] \quad (6)$$

And then the only possible angle for f that allows transits is $f = \pi/2$.

$$\cos^2(\pi/2) + \cos^2 I \sin^2(\pi/2) \leq \frac{R_*^2}{a} \longrightarrow \cos^2 I \leq \frac{R_*^2}{a^2} = |\cos I| \leq \frac{R_*}{a} \quad (7)$$

c) [1 point] Calculate the probability for a transit to take place assuming an isotropic distribution ($\cos I \sim U[-1, 1]$). Express as a function of $r(f)$ (i.e., do not assume a circular orbit).

We know that $r(f)$ is given by:

$$r(f) = \frac{a(1 - e^2)}{1 + e \cos f} \quad (8)$$

Then the transit probability for the isotropic distribution with non circular orbit is given by:

$$P_T = \frac{A_T}{A_a} = \frac{4\pi a R_*}{4\pi a^2} \frac{(1 + e \cos f)}{(1 - e^2)} \quad (9)$$

$$P_T(f) = \frac{R_*(1 + e \cos f)}{a(1 - e^2)} \quad (10)$$

d) [2 point] Call $\mathcal{P}(f)$ the probability in c. Compute $\langle \mathcal{P} \rangle_f$ and express it terms of the orbital angular momentum (or the semi-latus rectum).

From Class 4 we know that the Keplerian average of a function is given by:

$$\langle \phi(f) \rangle = \frac{(1 - e^2)^{3/2}}{2\pi} \int_0^{2\pi} \frac{df}{(1 + e \cos f)^2} \phi(f) \quad (11)$$

Using the probability recently calculated and solving the integral using Wolfram, we get:

$$\langle P(f) \rangle = \frac{(1 - e^2)^{3/2}}{2\pi} \int_0^{2\pi} \frac{df}{(1 + e \cos f)^2} \frac{R_*(1 + e \cos f)}{a(1 - e^2)} = \frac{R_*}{a} \quad (12)$$

In order to express the Keplerian average of the probability in terms of the orbital angular momentum we can use the expression seen in Class 4, given by;

$$L = \sqrt{GMa} \longrightarrow a = \frac{L^2}{GM} \quad (13)$$

Finally we can write $\langle \mathcal{P} \rangle_f$ as:

$$\langle \mathcal{P} \rangle_f = \frac{GMR_*}{L^2} \quad (14)$$

Problem 2: potential from an oblate star [3 points]

Let us assume that a planet orbits an oblate star and we can write the potential at quadrupolar order as:

$$V(r) = -\frac{GM_\star}{r} \left[1 - J_2 \left(\frac{R_s}{r} \right)^2 \right] \quad (15)$$

where $J_2 \ll 1$ is dimensionless¹. Note that this is only valid for zero-inclination orbits (relative to the stellar equator)

a) [1 point] Orbit-average the contribution term from the oblateness ($\propto 1/r^3$)

We can rewrite the potential as:

$$V(r) = -GM_\star \left[\frac{1}{r} - J_2 \frac{R_s^2}{r^3} \right] \quad (16)$$

Then using the previously shown expression for $r(f)$, we can write the whole potential expression as:

$$V(r) = -GM_\star \left[\frac{1 + e \cos f}{a(1 - e^2)} - J_2 R_s^2 \left(\frac{1 + e \cos f}{a(1 - e^2)} \right)^3 \right] \quad (17)$$

If we only consider the terms that are multiplied by the constant J_2 (X_{oblate}), we can use the Keplerian function average to evaluate the integral in Wolfram. The result is given by:

$$\langle X_{oblate}(f) \rangle = \frac{(1 - e^2)^{3/2}}{2\pi} \int_0^{2\pi} \frac{df}{(1 + e \cos f)^2} J_2 GM_\star R_s^2 \left(\frac{1 + e \cos f}{a(1 - e^2)} \right)^3 = \frac{J_2 GM_\star R_s^2}{a^3 (1 - e^2)^{3/2}}$$

b) [2 point] Right down the full Hamiltonian in terms of the Delaunay variables.

The Hamiltonian is given by:

$$H = K + V \quad (18)$$

Where K and V correspond to kinetic and potential energy components. We can write the kinetic component as a function of f as it is shown in the following equation:

$$K = \frac{GM}{2} \left(\frac{2(1 + e \cos f)}{a(1 - e^2)} - \frac{1}{a} \right) \quad (19)$$

Now we can write the Hamiltonian as a function of f , and then we can apply Keplerian average.

$$\langle H \rangle = \langle K \rangle + \langle V \rangle \quad (20)$$

Solving each integral in Wolfram we get

$$\langle H \rangle = \frac{GM}{2a} + \frac{J_2 GM_\star R_s^2}{a^3 (1 - e^2)^{3/2}} - \frac{GM_\star}{a} = -\frac{GM}{2a} + \frac{J_2 GM R_s^2}{a^3 (1 - e^2)^{3/2}} \quad (21)$$

¹Typically stars have $J_2 \sim 10^{-2} - 10^{-4}$ early on when they rotate fast. It becomes smaller for slowly-rotating star

Finally we can replace using Delaunay variables expressions.

$$\langle H \rangle = -\frac{GM}{2} \left(\frac{GM}{L^2} \right) + J_2 GM R_s^2 \left(\frac{GM}{L^2} \right)^3 \left(\frac{L}{\mathcal{G}} \right)^3 = -\frac{(GM)^2}{2L^2} + \frac{J_2 R_s^2 (GM)^4 L}{\mathcal{G}^3} \quad (22)$$

Problem 3: a hot Jupiter with an outer companion [5 points]

Estos últimos 2 ejercicios los tuve que hacer con nada de tiempo, así que probablemente no estén muy buenos

Consider the two-planet system HAT-P-13 ([link](#)).

a) [1 point] Compute the probability that planet b transits. Repeat for c.

For isotropic and independent transits, the probability is given by:

$$P_t = \frac{R_*}{a} \quad (23)$$

HAT-P-13 radius is given by $1.73_{-0.09}^{+0.10} R_\odot$ and b's semimajor axis is given by $0.04313_{-0.00026}^{+0.00033}$ AU. Replacing both values on Equation 23, we get:

$$P_t(b) = 0.18653625 \quad (24)$$

Similarly for c, with a semimajor axis of $1.188_{-0.33}^{+0.018}$ AU, the probability is given by:

$$P_t(c) = 0.00677215 \quad (25)$$

b) [2 point] Compute the probability that both planet b and planet c transit in two limits: i) isotropic and independent inclinations; ii) coplanar inclinations.

i) We know that for isotropic and independent inclinations, the probability that both planets transit, is given by:

$$P_t(b \wedge c) = \frac{R_*}{a_b} \times \frac{R_*}{a_c} \quad (26)$$

Then, using Equation 26, we get that for isotropic and independent inclinations the probability that both planets transit is equal to:

$$P_t(b \wedge c) = 0.00126325 \quad (27)$$

ii) For coplanar inclinations we have that $\cos I_b = \cos I_c$. The probability that both planet b and planet c transit their star depends on their orbital parameters, like their semi-major axes and the orientation of their orbits relative to the observer. If they have coplanar inclinations, it means that their orbital planes are aligned in such a way that they can both potentially transit the star when observed from a certain vantage point. In order to compute a probability, more information is needed.

c) [2 points] Assume that both planets lie in the x-y plane (zero inclinations). Integrate the system using Rebound and calculate the radial velocity of the system for a baseline of 3 years. You may assume that the observer is at $\hat{\mathbf{x}}$. Show the results for planets b and c separately, as well as both planets. How does the amplitude compare with the analytical estimates?

Results are shown in Figure 1.

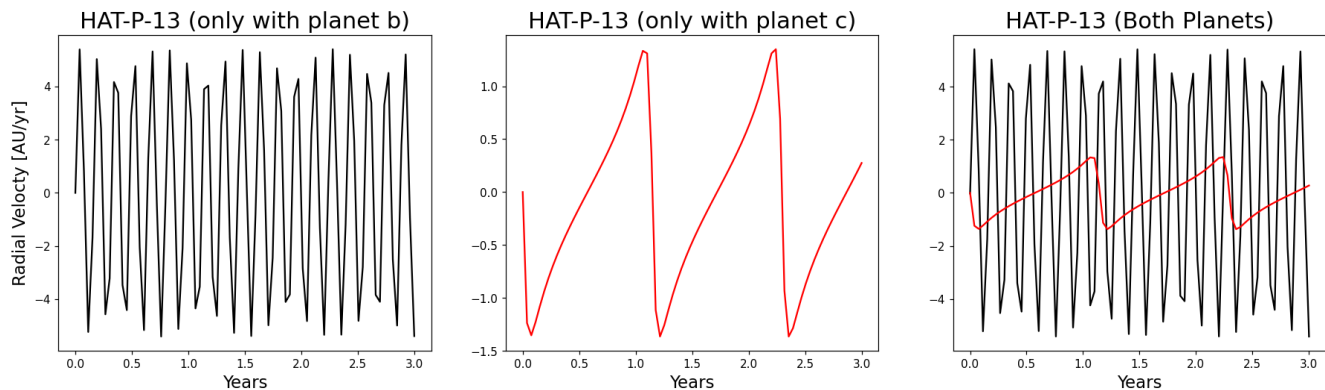


Figure 1: Simulated radial velocities for HAT-P-13

Problem 4: dynamics of a two-planet system [3 points]

Using HAT-P-13 from problem 2, we will explore the dynamics of a coplanar two-planet system and see when the secular evolution holds.

a) [2 point] Integrate the system for 100 years setting $f = \Omega = I = 0$, $\omega = \pi/2$, and plot a_b and e_b . Is the evolution conserving orbital energy of planet b?, what happens to its eccentricity?

Figure 2 shows the semi-major axis and eccentricity evolution.

We can approximate the energy of the planet as the expression shown below.

$$E = \frac{GMm}{2a} \quad (28)$$

Where a corresponds to the semi-major axis. We can see that even though the semi-major axis values is not perfectly constant, it keeps its values within a given range. So we can consider that the energy is conserved. We can see that the eccentricity has an increasing trend over time.

b) [1 point] Repeat the exercise by placing planet b at $a_b = 0.2$ au. How does the result change?

We can see that if we set the semi-major axis of planet b to 2 AU, after approximately 20 years the a value starts heavily oscillating and the eccentricity starts getting way bigger (see the comparison between the y-axis in Figure 2 and 3). So it is possible that the system may lose equilibrium.

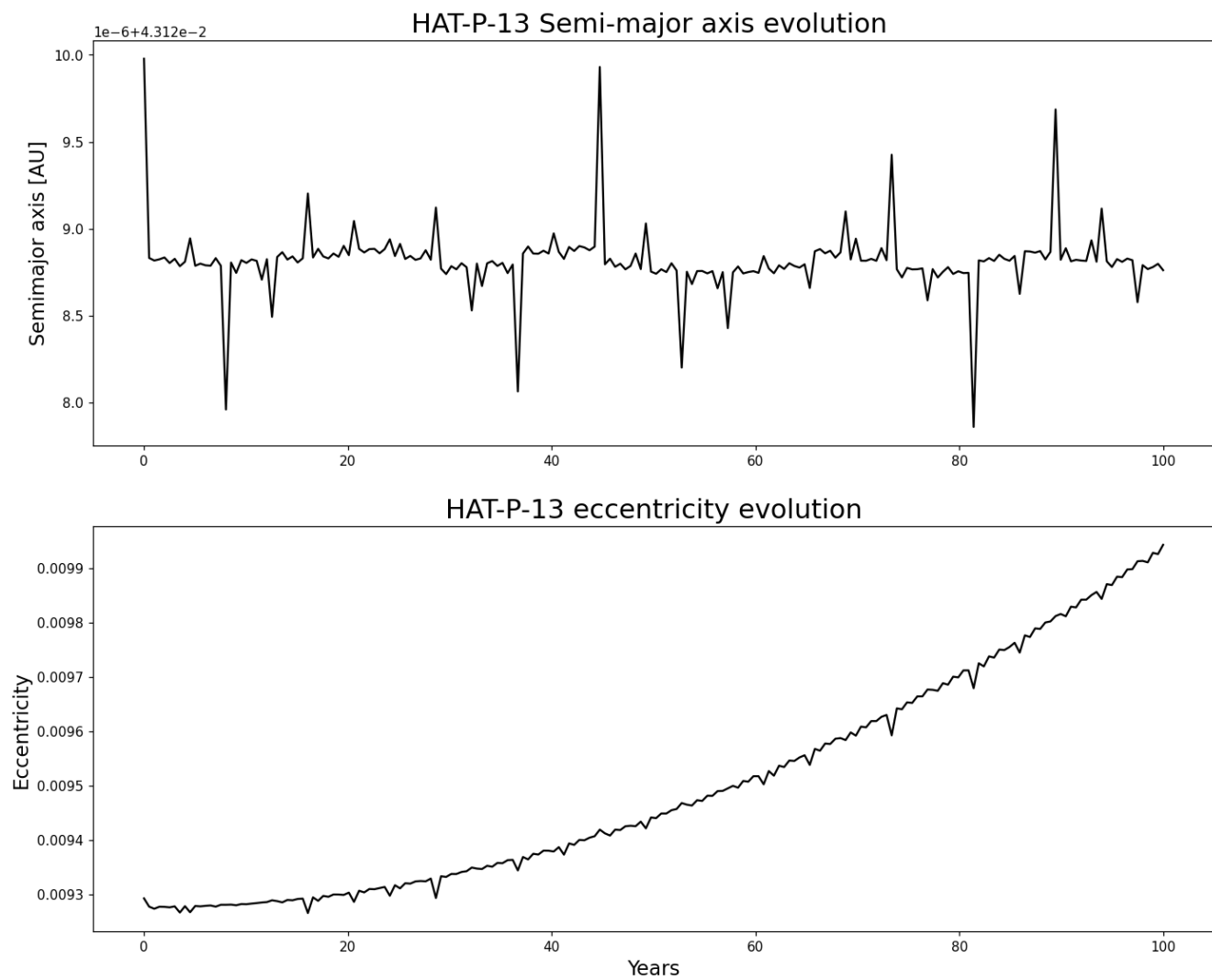


Figure 2: Semi-major axis (up) and eccentricity (down) evolution.

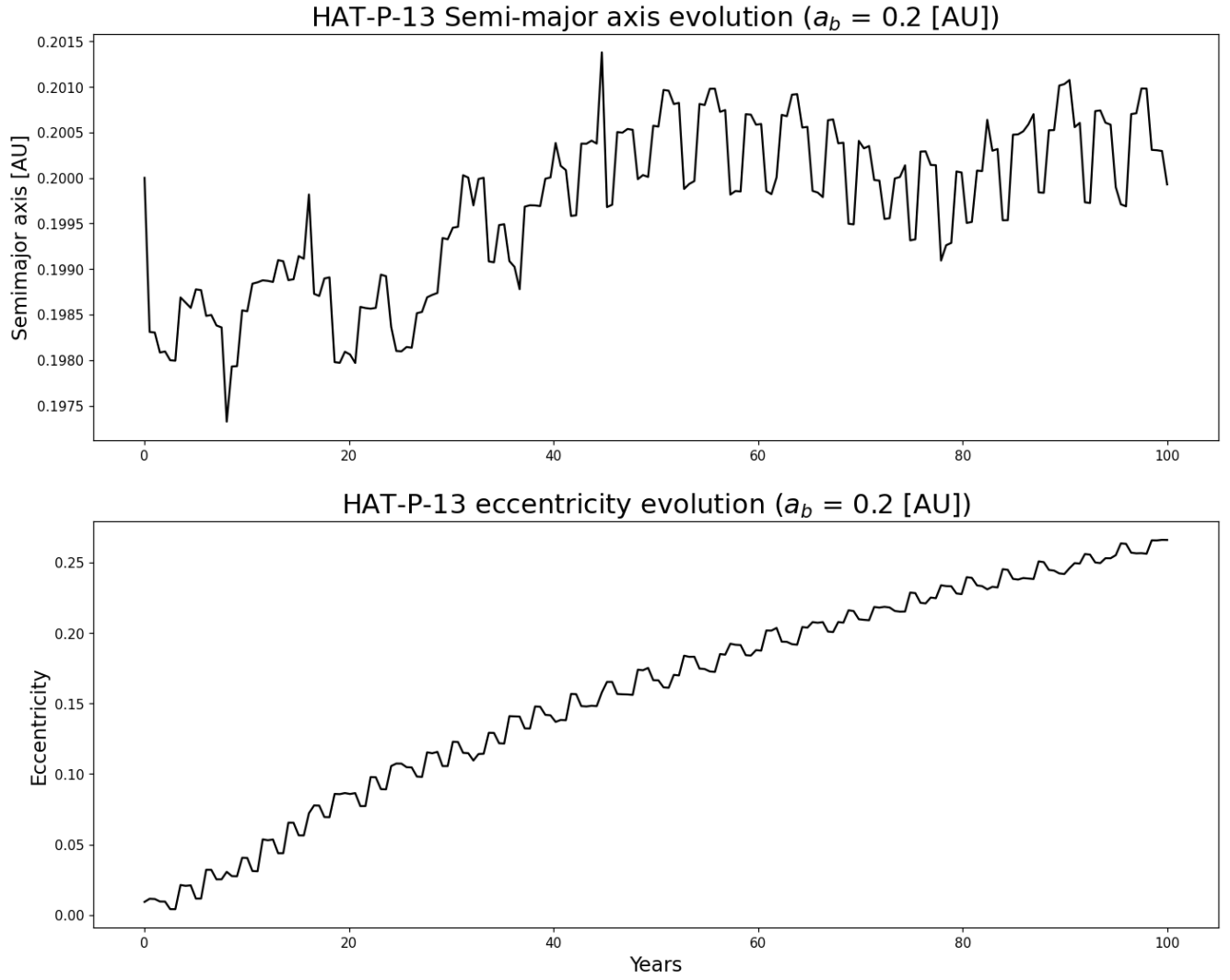


Figure 3: Semi-major axis (up) and eccentricity (down) evolution for $a_b = 2$ [AU]